Technical Report on YOLO Dataset Meta-Configuration Files

Abstract

This report presents a structured overview of metadata configuration files used in YOLO-based object detection datasets. These files define dataset organization, class mappings, annotation formats, and training inputs required for effective model learning.

1. YAML Configuration File

The YAML file acts as the central configuration file that defines dataset paths and class details.

```
train: images/train val: images/val nc: 3 names: ['person', 'car',
'bike']
```

train: Path to training images
val: Path to validation images
nc: Number of classes
names: Class labels

2. Dataset Directory Structure

```
dataset/ ■■■ images/ ■ ■■■ train/ ■ ■■■ val/ ■ ■■■ test/ ■■■ labels/
■ ■■■ train/ ■ ■■■ val/ ■ ■■■ test/ ■■■ data.yaml
```

Each image has a corresponding label file stored in the labels directory.

3. Label Annotation Format

```
0 0.52 0.43 0.30 0.25
```

Format: **class_id x_center y_center width height**
All values are normalized between 0 and 1 relative to image dimensions.

4. train.txt and val.txt

```
images/train/img_001.jpg images/train/img_002.jpg
```

These files list the image paths used during training and validation.

5. Dataset Creation Strategy

From a 1-minute video sampled at 10 FPS, approximately 600 images are generated. A subset of images is selected to create training, validation, and test datasets.

Training: 100 images
Validation: 40 images
Test: Remaining images

6. Labeling Tool

Label Studio is used to annotate images by drawing bounding boxes and assigning class labels. Accurate labeling is critical for achieving reliable model performance.

7. Importance of Metadata

Metadata files guide the training process by defining dataset structure, class mappings, and annotation formats. Incorrect configurations can lead to training failures or reduced accuracy.

8. References

- Ultralytics Documentation: https://docs.ultralytics.com/datasets/detect/
- Label Studio Documentation: https://labelstud.io/
- YOLO Official Documentation